Mode 3 — Square

DP8 (16×8) → DP8 (16×4) → DP8 (8×4) → DP8 (4×4) → add-and-square

Step 1 — DP8 (16×8)

$$\mathrm{DP8}\,(16{\times}8) = \sum_{i=0}^{7} a_i\, b_i$$

Step 2 — DP8 (16×4)

$$\mathrm{DP8}\,(16{\times}8) = 2^4 \underbrace{\sum_{i=0}^{7} a_i\, b_i^{hi}}_{\mathrm{DP8}\,(16\times4)} + \underbrace{\sum_{i=0}^{7} a_i\, b_i^{lo}}_{\mathrm{DP8}\,(16\times4)}$$

Step 3 — DP8 (8×4)

$$\mathrm{DP8}\,(16{\times}8) = 2^{12} \underbrace{\sum_{i=0}^{7} a_i^{hi}\, b_i^{hi}}_{\mathrm{DP8}\,(8\times4)} + 2^{8} \underbrace{\sum_{i=0}^{7} a_i^{hi}\, b_i^{lo}}_{\mathrm{DP8}\,(8\times4)} + 2^{4} \underbrace{\sum_{i=0}^{7} a_i^{lo}\, b_i^{hi}}_{\mathrm{DP8}\,(8\times4)} + \underbrace{\sum_{i=0}^{7} a_i^{lo}\, b_i^{lo}}_{\mathrm{DP8}\,(8\times4)}$$

Step 4 — DP8 (4×4)

$$\begin{aligned}\mathrm{DP8}\,(16{\times}8) = {} & 2^{16} \underbrace{\sum_{i=0}^{7} a_{H,i}^{hi}\, b_i^{hi}}_{\mathrm{DP8}\,(4\times4)} + 2^{12} \underbrace{\sum_{i=0}^{7} a_{L,i}^{hi}\, b_i^{hi}}_{\mathrm{DP8}\,(4\times4)} + 2^{12} \underbrace{\sum_{i=0}^{7} a_{H,i}^{hi}\, b_i^{lo}}_{\mathrm{DP8}\,(4\times4)} + 2^{8} \underbrace{\sum_{i=0}^{7} a_{L,i}^{hi}\, b_i^{lo}}_{\mathrm{DP8}\,(4\times4)} \\ & + 2^{8} \underbrace{\sum_{i=0}^{7} a_{H,i}^{lo}\, b_i^{hi}}_{\mathrm{DP8}\,(4\times4)} + 2^{4} \underbrace{\sum_{i=0}^{7} a_{L,i}^{lo}\, b_i^{hi}}_{\mathrm{DP8}\,(4\times4)} + 2^{4} \underbrace{\sum_{i=0}^{7} a_{H,i}^{lo}\, b_i^{lo}}_{\mathrm{DP8}\,(4\times4)} + \underbrace{\sum_{i=0}^{7} a_{L,i}^{lo}\, b_i^{lo}}_{\mathrm{DP8}\,(4\times4)}\end{aligned}$$

Step 5 — Add-and-square (substitute each DP8 (4×4) $= \frac{1}{2}\left[\sum(x_i{+}b_i)^2 - \sum x_i^2 - \sum b_i^2\right]$)

$$\begin{aligned}\mathrm{DP8}\,(16{\times}8) = {} & 2^{16} \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}(a_{H,i}^{hi}{+}b_i^{hi})^2 - \sum_{i=0}^{7}(a_{H,i}^{hi})^2 - \sum_{i=0}^{7}(b_i^{hi})^2\Big)}_{\mathrm{DP8}\,(4\times4)} \\ & + 2^{12} \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}(a_{L,i}^{hi}{+}b_i^{hi})^2 - \sum_{i=0}^{7}(a_{L,i}^{hi})^2 - \sum_{i=0}^{7}(b_i^{hi})^2\Big)}_{\mathrm{DP8}\,(4\times4)} \\ & + 2^{12} \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}(a_{H,i}^{hi}{+}b_i^{lo})^2 - \sum_{i=0}^{7}(a_{H,i}^{hi})^2 - \sum_{i=0}^{7}(b_i^{lo})^2\Big)}_{\mathrm{DP8}\,(4\times4)} \\ & + 2^{8} \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}(a_{L,i}^{hi}{+}b_i^{lo})^2 - \sum_{i=0}^{7}(a_{L,i}^{hi})^2 - \sum_{i=0}^{7}(b_i^{lo})^2\Big)}_{\mathrm{DP8}\,(4\times4)} \\ & + 2^{8} \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}(a_{H,i}^{lo}{+}b_i^{hi})^2 - \sum_{i=0}^{7}(a_{H,i}^{lo})^2 - \sum_{i=0}^{7}(b_i^{hi})^2\Big)}_{\mathrm{DP8}\,(4\times4)}\end{aligned}$$

$$+\, 2^4 \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}(a^{lo}_{L,i}{+}b^{hi}_i)^2 - \sum_{i=0}^{7}(a^{lo}_{L,i})^2 - \sum_{i=0}^{7}(b^{hi}_i)^2 \Big)}_{\text{DP8 (4×4)}}$$

$$+\, 2^4 \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}(a^{lo}_{H,i}{+}b^{lo}_i)^2 - \sum_{i=0}^{7}(a^{lo}_{H,i})^2 - \sum_{i=0}^{7}(b^{lo}_i)^2 \Big)}_{\text{DP8 (4×4)}}$$

$$+\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}(a^{lo}_{L,i}{+}b^{lo}_i)^2 - \sum_{i=0}^{7}(a^{lo}_{L,i})^2 - \sum_{i=0}^{7}(b^{lo}_i)^2 \Big)}_{\text{DP8 (4×4)}}$$

Step 6 — All components

$$\boxed{\begin{aligned}
\mathrm{PE} &= 2^{16}\sum_{i=0}^{7}(a^{hi}_{H,i}{+}b^{hi}_i)^2 + 2^{12}\sum_{i=0}^{7}(a^{hi}_{L,i}{+}b^{hi}_i)^2 + 2^{12}\sum_{i=0}^{7}(a^{hi}_{H,i}{+}b^{lo}_i)^2 + 2^{8}\sum_{i=0}^{7}(a^{hi}_{L,i}{+}b^{lo}_i)^2 \\
&\quad + 2^{8}\sum_{i=0}^{7}(a^{lo}_{H,i}{+}b^{hi}_i)^2 + 2^{4}\sum_{i=0}^{7}(a^{lo}_{L,i}{+}b^{hi}_i)^2 + 2^{4}\sum_{i=0}^{7}(a^{lo}_{H,i}{+}b^{lo}_i)^2 + \sum_{i=0}^{7}(a^{lo}_{L,i}{+}b^{lo}_i)^2 \\[1ex]
\alpha &= 2^{16}\sum_{i=0}^{7}(a^{hi}_{H,i})^2 + 2^{12}\sum_{i=0}^{7}(a^{hi}_{L,i})^2 + 2^{12}\sum_{i=0}^{7}(a^{hi}_{H,i})^2 + 2^{8}\sum_{i=0}^{7}(a^{hi}_{L,i})^2 \\
&\quad + 2^{8}\sum_{i=0}^{7}(a^{lo}_{H,i})^2 + 2^{4}\sum_{i=0}^{7}(a^{lo}_{L,i})^2 + 2^{4}\sum_{i=0}^{7}(a^{lo}_{H,i})^2 + \sum_{i=0}^{7}(a^{lo}_{L,i})^2 \\[1ex]
\beta &= 2^{16}\sum_{i=0}^{7}(b^{hi}_i)^2 + 2^{12}\sum_{i=0}^{7}(b^{hi}_i)^2 + 2^{12}\sum_{i=0}^{7}(b^{lo}_i)^2 + 2^{8}\sum_{i=0}^{7}(b^{lo}_i)^2 \\
&\quad + 2^{8}\sum_{i=0}^{7}(b^{hi}_i)^2 + 2^{4}\sum_{i=0}^{7}(b^{hi}_i)^2 + 2^{4}\sum_{i=0}^{7}(b^{lo}_i)^2 + \sum_{i=0}^{7}(b^{lo}_i)^2
\end{aligned}}$$